



Course Details

Subject {L-T-P / C} : CS6218 : Machine Learning {3-0-0 / 3}

Subject Nature : Theory

Coordinator : Puneet Kumar Jain

Syllabus

- Module 1:** Introduction and regression
Machine learning introduction with major milestone and motivating applications.
Recap of basic mathematics concepts (Linear algebra, Matrix calculus and Probability and statistics).
Linear regression: ridge, Lasso
- Module 2:** Supervised classification
Logistic regression, Gradient Descent, Perceptron to Neuron, ANN, Backpropagation, Evaluation metrics, Cross validation, Bias-Variance trade off, regularisation.
Naïve Bayes, Linear and Gaussian Discriminant analysis, Kernel methods and Support vector machine, KNN
Decision Tree, Ensemble methods; bagging, boosting, and random forest
- Module 3:** Unsupervised Learning
Clustering: K-means, k-medoids, Hierarchical clustering, Agglomerative, DBScan, spectral clustering
- Module 4:** Reinforcement learning and miscellaneous topics
Reinforcement learning, Mixture of Gaussians (GMM), Expectation maximization, PCA and ICA, Bayesian Networks, Intro to Semi-supervised Learning

Course Objective

1. To introduce the fundamental principles and mathematical foundations of machine learning, including linear algebra, probability, and regression techniques.
2. To equip students with supervised learning methods such as neural networks, support vector machines, and ensemble techniques for solving classification problems.
3. To familiarize students with unsupervised learning algorithms for clustering, pattern discovery, and dimensionality reduction.
4. To expose students to advanced and emerging topics like reinforcement learning, probabilistic models, and semi-supervised learning for real-world applications.

Course Outcome

1. Students will be able to understand core concepts and mathematical foundations of machine learning, including linear regression and regularization techniques.
2. Students will be able to apply supervised learning algorithms such as logistic regression, neural networks, and ensemble methods for classification tasks.
3. Students will be able to evaluate and optimize model performance using metrics, cross-validation, bias-variance trade-off, and regularization techniques.
4. Students will be able to implement unsupervised learning techniques like clustering and dimensionality reduction for exploratory data analysis.
5. Students will be able to explore advanced learning paradigms including reinforcement learning, probabilistic models, and semi-supervised learning approaches.

Essential Reading

1. Tom Mitchell, Machine Learning, McGraw Hill , 1997, ISBN 0-07-042807-7
2. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer , 2011 edition

Supplementary Reading

1. Kevin Murphy, Machine Learning: A Probabilistic Perspective, The MIT Press , 2012
2. Ian Goodfellow, Yoshua Bengio, Aaron Courville, Deep Learning, MIT Press , 2016

3. Richard O. Duda, Peter E. Hart, David G. Stork, Pattern classification, Wiley , (2nd edition). Wiley, New York, 2001

Journal and Conferences

1. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
2. Journal of Machine Learning Research (JMLR)
3. Neural Information Processing Systems (NeurIPS)
4. International Conference on Machine Learning (ICML)
5. Conference on Computer Vision and Pattern Recognition (CVPR)

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